Ridham Sonani

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Summary

Software engineer with 2+ years of experience designing and deploying full-stack applications and serverless cloud solutions. Skilled in Python, TypeScript, and AWS services, with a focus on building performant REST APIs, integrating AI/ML capabilities, and improving system reliability through observability and automation.

Education

Arizona State University Aug 2024 – May 2026
Master's, Computer Science | GPA: 3.77/4.0

Dhirubhai Ambani Institute of Information and Communication Technology Aug 2019 – Jun 2023
BTech, Information and Communication Technology

- **Coursework:** DSA, Operating Systems, DBMS, Software Engineering, Cloud Computing | **Certification:** AWS Certified AI Practitioner

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, C, SQL
Web Development: React.js, Next.js, Node.js, Express.js, REST APIs, Spring Boot, HTML5, CSS3, Material UI
Cloud & DevOps: AWS (Lambda, S3, API Gateway, Bedrock, Textract, CDK, Amplify, EC2), Docker, Git, CI/CD Pipelines, Infrastructure as Code
Databases: PostgreSQL, MySQL, DynamoDB
Tools & Platforms: Linux, GitHub, Postman, CloudWatch, Jira
Areas of Expertise: Cloud Architecture, Serverless Development, Generative AI Integration, Full-Stack Development, SDLC

Experience

Associate Cloud Developer, AWS Cloud Innovation Center – *Tempe, AZ* Jan 2025 – May 2026

- Architected cloud-native features using Lambda and API Gateway for applications serving 1,000+ users, owning full lifecycle from requirements through production.
- Developed Python REST APIs with structured logging and CloudWatch monitoring, automating incident response for improved operational efficiency.
- Integrated Generative AI services (Bedrock, Q Business) into customer-facing prototypes, enabling intelligent document processing and improved user interactions.

Full Stack Developer, Escrow Infotech – *Surat, India* Jan 2024 – Jul 2024

- Developed and optimized Python/Node.js backend APIs, reducing response latency from 450ms to 300ms for 500+ daily users.
- Enhanced PostgreSQL performance through indexing strategies and query tuning, cutting heavy query execution times by 35%.
- Crafted responsive UIs using HTML5, CSS3, and JavaScript, improving page load times by 20%.

Software Development Engineer Intern, Amazon – *Bangalore, India* Jan 2023 – Jun 2023

- Launched QA Metrics Portal automating test execution and defect metrics collection, saving program managers 8–10 hours weekly.
- Automated 200+ test cases on Echo devices using Python test runner simulating voice commands and UI validation, reducing testing effort by 70%.
- Developed Python test framework with reusable modules for voice command simulation and UI state verification, improving test reliability.

Projects

Historical Water Map Digitization System Jan 2025 – Mar 2025

- Engineered event-driven AWS pipeline (S3 triggers + Lambda) automating OCR extraction, LLM-based metadata identification, and geocoding at scale.
- Integrated AWS Textract and Bedrock (Claude 3.5) with prompt engineering to extract water resource entities from noisy OCR text.
- Deployed Infrastructure as Code via AWS CDK with CloudWatch monitoring and automated GeoJSON delivery to GitHub.

Technologies: Python, TypeScript, AWS (Lambda, S3, Textract, Bedrock, IAM, CDK), GitHub API, Geopy

AI Assistant Platform Mar 2025 – May 2025

- Shipped full-stack AI chatbot with React frontend and Python/Lambda backend, integrating Amazon Q Business for natural language search over indexed web and document sources.
- Configured knowledge ingestion pipeline with S3-backed data sources and EventBridge sync, delivering source-attributed responses grounded in indexed content.
- Automated deployment using AWS CDK and CodeBuild CI/CD, provisioning 10+ AWS resources with one-command scripts.

Technologies: React, TypeScript, Python, AWS (Lambda, S3, Q Business, Amplify, API Gateway, CDK), Material-UI